



# Unlocking implicit motion for evaluating image complexity<sup>☆</sup>

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## ABSTRACT

Image complexity (IC) plays a critical role in both cognitive science and multimedia computing, influencing visual aesthetics, emotional responses, and tasks such as image classification and enhancement. However, defining and quantifying IC remains challenging due to its multifaceted nature, which encompasses both objective attributes (e.g., detail, structure) and subjective human perception. While traditional methods rely on entropy-based or multidimensional approaches, and recent advances employ machine learning and shallow neural networks, these techniques often fail to fully capture the subjective aspects of IC. Inspired by the fact that the human visual system inherently perceives implicit motion in static images, we propose a novel approach to address this gap by explicitly incorporating hidden motion into IC assessment. We introduce the motion-inspired image complexity assessment metric (MICM) as a new framework for this purpose. MICM introduces a dual-branch architecture: One branch extracts spatial features from static images, while the other generates short video sequences to analyze latent motion dynamics. To ensure meaningful motion representation, we design a hierarchical loss function that aligns video features with text prompts derived from image-to-text models, refining motion semantics at both local (i.e., frame and word) and global levels. Experiments on three public image complexity assessment (ICA) databases demonstrate that our approach, MICM, significantly outperforms state-of-the-art methods, validating its effectiveness. The code will be publicly available upon acceptance of the paper.

## 1. Introduction

The study of image complexity (IC) is significant in both cognitive science [1,2] and multimedia computing [3,4]. In psychology, IC influences visual aesthetics [4,5] and emotional reactions [6], while in computer vision, it affects diverse tasks. Automating image complexity assessment (ICA) has shown value in applications including fire detection [3], image steganography [7], interactive image segmentation [8], aesthetic image classification [9], and image enhancement [10]. Precisely quantifying IC bridges the human perception and computational modeling gap, enhancing downstream task performance and making it essential in perceptual research and computational approaches. However, defining IC remains challenging due to its multifaceted nature. Objectively, IC reflects the level of detail, diversity, and structural intricacy within an image. In existing research [11], image complexity is defined as an attribute reflecting the number, dimensions, and visibility of image details. In IC9600 [12], image complexity is defined as the intricacy contained within an image. Subjectively, it represents

the cognitive challenge that a human observer faces in interpreting or describing an image [1,2].

In the early stages of image complexity research, proposals were limited but diverse. Foundational studies often approached image complexity based on entropy. For instance, Stamps et al. [13] investigated the relationship between subjective visual diversity, pleasure, and statistical entropy in structured stimuli, revealing a strong linear correlation between perceived diversity and entropy. Rosenholtz et al. [14] explored visual clutter in user interfaces using three computational measures: feature congestion, subband entropy, and edge density. These measures were found to correlate with set size in visual search models and impact search performance. Cardaci et al. [15] proposed a fuzzy mathematical model based on fuzzy entropy measures, further expanding the entropy-based approach.

Meanwhile, other studies adopted multidimensional perspectives. Olivia et al. [16] argued that visual complexity is better represented by a multidimensional space rather than a single perceptual dimension.

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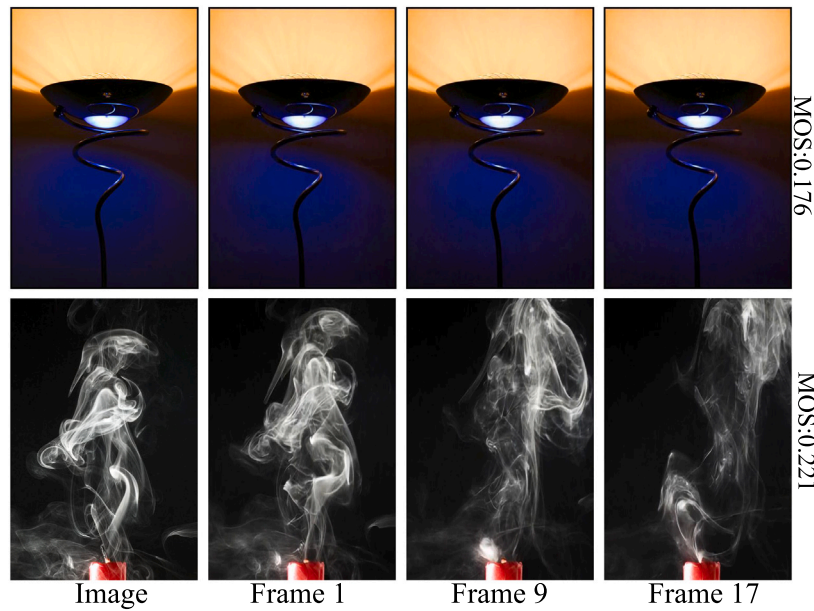


Fig. 1. Examples of images in IC9600 database. The **Frame 1/9/17** is sampled from the corresponding generated short videos. The applied Image-to-video (I2V) model is CogVideoX [25].

They identified key dimensions such as object quantity, clutter, openness, symmetry, organization, and color variety through hierarchical grouping tasks. Forsythe et al. [17] evaluated various visual complexity measures, noting potential biases in subjective judgments due to familiarity. Purchase et al. [18] examined the alignment between objective quantification and human perception of visual complexity, finding that while subjective judgments were consistent across individuals, they did not directly correlate with common computational metrics. Yu et al. [19] explored compression-based measures, demonstrating a strong correlation between spatial information (SI) and compression metrics and noting that compression-based complexity typically increases as resolution decreases.

Nevertheless, the above-mentioned early methods have shortcomings, particularly in their ability to fully capture the subjective nature of human perception and the multifaceted aspects of visual complexity. Recent studies have increasingly adopted machine learning and deep learning for ICA. Machado et al. [20] used edge density and compression error in machine learning models to predict visual complexity. Guo et al. [21] employed SVMs to evaluate painting complexity based on human perception and low-level features. Chen et al. [22] proposed a neural network method focusing on texture, edges, and significant areas. Abdelwahab et al. [23] utilized CNNs to enhance complexity assessment, while Saraee et al. [24] extracted unsupervised features from intermediate CNN layers, revealing their relevance to complexity. Feng et al. [12] introduced a dual-branch architecture, combining high-resolution detail and downsampled context features.

However, these methods have limitations: shallow models lack sufficient learning capacity, and other approaches focus only on local features in static images, neglecting broader contextual or dynamic aspects. An ensuing question arises: **Does the static information of an image sufficiently correspond to its perceived complexity? Specifically, does the implicit motion information within a static image influence the subjective assessment?**

Psychological studies on implicit motion and representational momentum [26] suggest that motion is perceived in certain static images. Our visual system is inherently designed to detect motion, even when it is merely implied, and to anticipate potential outcomes of actions—even when those actions involve only the subjects depicted in static images. Elisabetta et al. [27] argue that people rely on spatial context,

specifically the relationships among objects, to infer actions and predict future states, even in the absence of directly perceived physical events. Andrea et al. [28] demonstrate that implied motion in static photographs can alter the perceived position of stationary objects. Kartik et al. [29] further highlight that humans can routinely deduce complex sequences of past and future events from a single static snapshot of a dynamic scene, even in unfamiliar situations. As illustrated in Fig. 1, when viewing images depicting motions (e.g., the second row), human observers may instinctively associate them with specific movements. In contrast, images without explicit actions (e.g., the first row) do not evoke any additional motion-related information. Building on these findings, we hypothesize that implicit motion cues play a crucial role in the ICA task.

Such perceptual sensitivity also underpins the relevance of image quality assessment (IQA), which is intrinsically connected to the ICA task. Recent IQA methods increasingly emphasize alignment with the human visual system. For example, TOPIQ [30] proposed a top-down framework that explicitly models perceptual characteristics, while RichIQA [31] extended traditional Mean Opinion Score approaches by introducing the Distribution and Standard Deviation of Opinion Scores, thus capturing more nuanced subjective feedback. Other methods, such as BPRI [32] and MBPRI [33], introduced pseudo-reference images to blind IQA by simulating full-reference settings, better reflecting human visual perception. Inspired by the multi-channel processing nature of the visual system, the CVC-IM module [34] was developed to extract compact and informative mutual representations across different views. Similarly, UCA [35] employed a multi-scale weighting strategy to integrate perceptual variations across content types and scales. Although IQA methods have been extensively studied, their perceptual focus differs from that of the ICA task, making the design of a well-grounded ICA framework still challenging.

On the other hand, some studies have explored multi-modal modeling. Specifically, UGC-New [36] established a new dataset specifically for User-Generated Content (UGC). Zhu et al. constructed both a visual attention dataset for Virtual Reality (VR) [37] and a saliency dataset for Augmented and Virtual Reality (AR/VR) [38], laying the groundwork for perceptual quality assessment across multi-modal content such as UGC and AR/VR videos. Wu et al. [39] constructed MMHead, which provides a large-scale benchmark for 3D facial animation incorporating

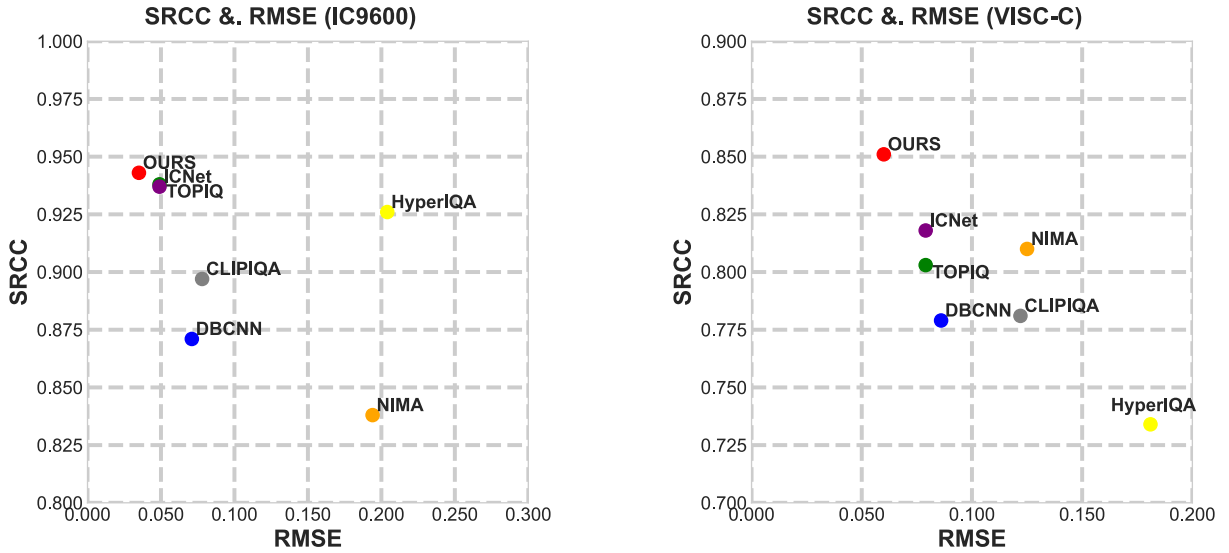


Fig. 2. The comparisons of SRCC $\uparrow$  and RMSE $\downarrow$  of proposed methods with state-of-the-art ICA/IQA approaches on IC9600 and VISC-C databases.

video, audio, and text modalities. Moreover, Min et al. adopted a multi-modal fusion strategy to study audiovisual quality prediction [40], combining auditory and visual cues to estimate human attention, especially in motion scenes [41]. Similarly, integrating auditory saliency to enhance predictions of visual saliency [42]. The VBAS [38] addressed the challenge of saliency prediction in augmented omnidirectional videos by extracting temporal and spatial visual cues and adopting different strategies to fuse spatial and temporal saliency. For a broader understanding of visual perception modeling and user experience, some overviews such as [43–47] provide valuable insights. However, the exploration of multi-modal design in the field of ICA is still limited.

Furthermore, the growing body of work on quality assessment for AI-generated content offers valuable insights into the challenge of controllability in AI-generated content. Several benchmarks have recently emerged: FineVD [48] provided a fine-grained video quality assessment dataset for user-generated content; AIGVQA-DB [49] focused on AI-generated videos; HarmonyIQAD [50] addresses image harmony quality; and AIGCIQA2023 [51] and its extension AIGCIQA2023+ [52] targeted image quality in AI-generated content. Further contributions include AIGC-T23DAQA [53], the largest quality dataset for text-to-3D assets, and FaceQ [54], which evaluated the quality of AI-generated face images based on fine-grained human preference annotations. Additionally, PrefIQA [55] introduced a cross-modal alignment and fusion framework to better assess the perceptual quality of generative content. The advancement of these AIGC quality assessment approaches has significantly improved the quality and consistency of generated content, paving the way for extracting latent motion information using generative models.

Motivated by these developments, we propose a novel metric — **motion inspired image complexity assessment metric (MICM)** — designed to capture latent motion information embedded within static images. Our approach adopts a dual-branch architecture: one branch processes spatial features from still images, while an auxiliary branch infers motion cues. Given that current ICA datasets lack corresponding video data, we synthesize short sequences using state-of-the-art image-to-text and image-to-video generation models. To ensure semantic relevance and controllability of the generated content, we introduce a hierarchical loss that aligns video features with textual prompts at both frame/word and global levels.

As shown in Fig. 2, our approach significantly outperforms state-of-the-art methods, demonstrating its effectiveness. The main contributions of this paper are as follows:

- (1) We propose a novel dual-branch framework, MICM, that combines spatial and motion information for ICA.
- (2) We integrate a hierarchical loss function to enhance motion representation in generated videos by cross-model alignments.
- (3) The empirical evidence highlights the importance of implicit motion in the human perception of IC.

The rest of this paper is divided into three sections. In Section 2, we introduce our proposed ICA approach, MICM, in detail. Section 3 shows the experimental results and analysis. Finally, we conclude the paper in Section 4.

## 2. Proposed method

We illustrate the framework of our proposed method in Fig. 3. Given an input image, we first generate a corresponding text prompt using a pretrained image-to-text (I2T) model. This text prompt and the image are then used to generate a short video sequence via an image-to-video (I2V) model. The generated videos are kept short (ranging from 16 to 49 frames, depending on the specific I2V model) to ensure content reliability and prevent the introduction of extraneous, artificially generated elements. Next, the image and its corresponding text-video pairs are processed through separate branches, ultimately leading to a combined complexity prediction. Further discussions about I2T and I2V models are presented in Section 3.5. The detailed design is as follows.

### 2.1. Image branch

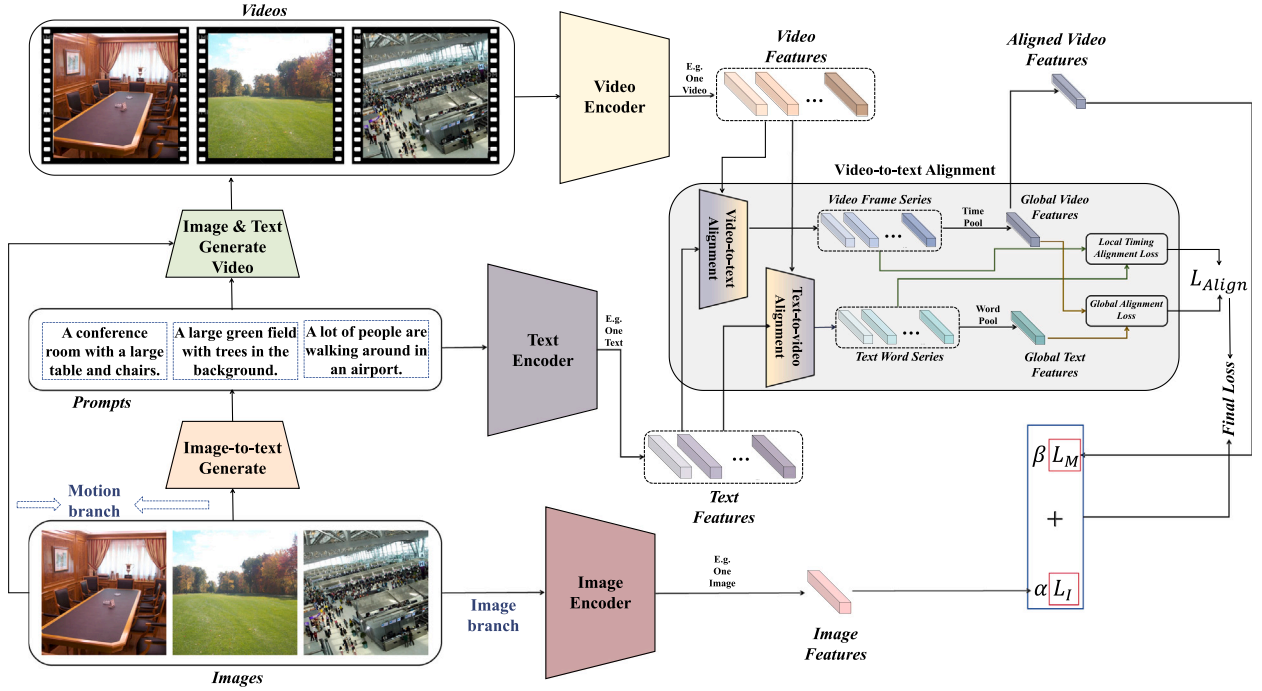
For each input image  $I \in \mathbb{R}^{3 \times W \times H}$ , where  $W$  and  $H$  denote the width and height, respectively, we directly employ a pretrained ResNet [56] as the image encoder to extract spatial features. The extracted feature representation  $F_I \in \mathbb{R}^{1 \times D}$  is subsequently passed through a linear regression layer to obtain the complexity prediction  $P_I$  for the image branch, where  $D$  is the embedding dimension. These processes can be formulated as follows:

$$F_I = F_I(I), \quad (1)$$

$$P_I = \text{Linear}(F_I), \quad (2)$$

$$L_I = \frac{1}{N} \sum_{i \in N} (P_I^i - S_I^i)^2. \quad (3)$$

Here,  $F_I(\cdot)$  represents the image encoder,  $\text{Linear}(\cdot)$  denotes the regression layer, and  $L_I$  is the mean squared error (MSE) loss function.  $N$  is the batch size, and  $S_I$  is the ground-truth complexity label.



**Fig. 3.** The overall framework of the proposed motion-inspired image complexity assessment metric (MICM) follows a dual-branch design. For example, we take one image and a corresponding generated text and video. The image branch captures spatial information from static images, contributing to the image loss  $L_I$ . Meanwhile, the motion branch extracts implicit motion information by aligning video features with text prompts at both local (i.e., frame and word) and global levels, resulting in the motion loss  $L_M$ . Additionally, a hierarchical loss  $L_{Align}$  is introduced to further enhance model training by ensuring consistency between the generated videos and texts. The three loss functions contribute to the final loss.

## 2.2. Motion branch

To incorporate implicit motion information, we first generate a text prompt  $T_I$  for each input image  $I$  using a pretrained image-to-text (I2T) model:

$$T_I = \text{I2T}(I). \quad (4)$$

We evaluate four state-of-the-art I2T models: BLIP [57], BLIP2 [58], InstructBLIP-Vicuna-7B [59], and InstructBLIP-Flan-T5-XL [59]. A detailed ablation study is presented in Section 3.5. Here, we provide examples of images in IC9600 and corresponding generated text prompts as shown in Fig. 4. It is evident that different I2T models generate text prompts with varying levels of detail. For images with implicit motion, the corresponding text consists of more explicit motion-related descriptions. For example, in the first row, the generated text includes words like “lit” and “burning” for the candle. In contrast, for images where motion information is less apparent, the text prompts primarily describe objects, scenes, or spatial relationships without incorporating verbs indicative of movement.

Next, we generate a short video sequence from the image-text pair using a pretrained image-to-video (I2V) model:

$$V_I = \text{I2V}(I, T_I), \quad V_I \in \mathbb{R}^{N_F \times 3 \times W \times H}, \quad (5)$$

where  $N_F$  represents the number of frames in the generated video. We evaluate three state-of-the-art I2V models: CogVideoX [25], CogVideoX1.5 [25], and VGen [60]. We also provide examples of images-text pairs and corresponding videos in Fig. 5. It is obvious that in terms of content coherence, videos generated by CogVideoX exhibit the highest quality. Specifically, the roller coaster video produced by CogVideoX shows minimal distortion, whereas the one generated by Vgen lacks continuity. Additionally, in the zebra grazing video, CogVideoX achieves a more natural depiction of tail and ear movements. Further ablation studies can be found in Section 3.5.

The generated video  $V_I$  and the corresponding text prompt  $T_I$  are then embedded using video and text encoders:

$$F_V = F_V(V_I), F_V \in \mathbb{R}^{N_F \times D} \quad (6)$$

$$F_T = F_T(T_I), F_T \in \mathbb{R}^{N_T \times D} \quad (7)$$

where  $F_T$  is a pretrained BERT model [61] that encodes textual features. Here,  $N_T$  denotes the number of tokens in the tokenized text, and  $D$  is the shared embedding dimension across modalities. The details of  $F_V$  are as follows:

$$F'_V = \text{ResNet}(V_I), F'_V \in \mathbb{R}^{N_F \times 4096}, \quad (8)$$

$$F_V = \text{GRU}(\text{Conv1d}(F'_V)), F_V \in \mathbb{R}^{N_F \times D}. \quad (9)$$

Here,  $F'_V$  represents the frame-wise spatial features extracted using a pretrained ResNet. A 1D convolution ( $\text{Conv1d}$ ) is then applied to reduce the feature dimension from 4096 to the embedding dimension  $D$ . Subsequently, a bidirectional GRU is employed for temporal modeling, enabling the extraction of motion information.

To ensure that the motion branch focuses more effectively on extracting motion-related information, we leverage the fact that text prompts inherently contain motion descriptions, as demonstrated in the study by Jae et al. [62]. To enhance the video features' sensitivity to motion cues, we introduce cross-modal alignment between video and text features, reinforcing the representation of motion dynamics.

As shown in Fig. 3, the alignment process begins with bidirectional multi-head attention. Specifically, the video-to-text attention is computed as

$$v_{\text{series}} = \text{MHA}_v(F_V, F_T, F_T), v_{\text{series}} \in \mathbb{R}^{N_F \times D} \quad (10)$$

where  $\text{MHA}_v(\cdot)$  is a multi-head attention mechanism (with four heads) that uses the video features as queries and the text features as both keys and values. Similarly, the text-to-video attention is defined by

$$t_{\text{series}} = \text{MHA}_t(F_T, F_V, F_V), t_{\text{series}} \in \mathbb{R}^{N_T \times D} \quad (11)$$



| Image-to-text<br>Generative Model                                                 | InstructBlip<br>Vicuna-7B                                                                           | InstructBlip<br>Flan-T5-XL                                                                                                            | Blip2<br>Opt-COCO-2.7B                          | Blip<br>Large-COCO                              |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
|  | A close up of a lit candle in a glass container.                                                    | The image depicts a candle in a glass container with a single burning candle inside.                                                  | A candle in a jar with a single lit candle.     | A candle is lit in a glass jar.                 |
|  | A close up view of some green leaves with water droplets on them.                                   | The image features a large group of green leaves in the foreground, along with several smaller leaves scattered throughout the scene. | A close up of a leaf with water droplets on it. | A close up of a leaf with water droplets on it. |
|  | A blurry image of fireworks in the night sky with blue, orange and red streaks across it's surface. | The image features a blurry background with multiple lights, creating a surreal and colorful scene.                                   | A blurry image of a city at night.              | A blurry image of a city at night.              |

Fig. 4. The examples of images in IC9600 database, and corresponding text prompts generated by BLIP [57], BLIP2 [58], and InstructBLIP [59] (Vicuna-7B, and Flan-T5-XL).

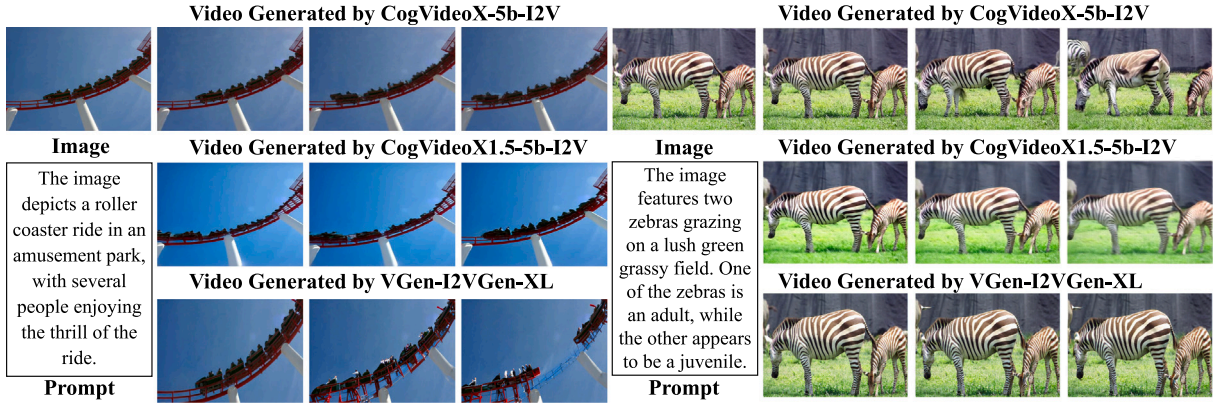


Fig. 5. The examples of images in IC9600 database, and corresponding text prompts generated by InstructBLIP [59] (Vicuna-7B). The video frames are sampled from three state-of-the-art image-to-video models (i.e., CogVideoX [25], CogVideoX1.5 [25], and Vgen [60].).

with  $\text{MHA}_t(\cdot)$  employing the text features as queries and the video features as keys and values.

To aggregate temporal information from the video, we introduce a learnable temporal weight vector  $w \in \mathbb{R}^{N_F}$ . After normalizing these weights with a softmax function, the global video feature  $v_{\text{global}} \in \mathbb{R}^D$  is computed by a weighted sum:

$$v_{\text{global}} = \text{softmax}(w)^\top \cdot v_{\text{series}}. \quad (12)$$

In parallel, the global text representation  $t_{\text{global}} \in \mathbb{R}^D$  is obtained by applying a linear projection to the first token (often regarded as the [CLS] token) of the attended text features:

$$t_{\text{global}} = \text{Linear}(t_{\text{series}}). \quad (13)$$

To achieve comprehensive alignment between video and text features, we propose a hierarchical loss that aligns both the global features  $v_{\text{global}}$  and  $t_{\text{global}}$ , as well as the sequential features  $v_{\text{series}}$  and  $t_{\text{series}}$  at the local (i.e., frame and word) level. The details of the hierarchical loss are as follows:

A learnable logit scale parameter  $\tau$  is used to adjust the similarity scores between video and text features before applying the contrastive loss. This parameter acts as a temperature parameter, controlling the sharpness of the softmax distribution over the similarity matrix, which in turn helps the model learn more discriminative and robust representations. The global similarity between  $v_{\text{global}}$  and  $t_{\text{global}}$  is computed as:

$$S_{\text{global}} = \tau v_{\text{global}} \cdot t_{\text{global}}^\top, \quad (14)$$

which is used within a contrastive loss framework. Specifically, the **global alignment loss** is defined as the InforNCE (Information Noise Contrastive Estimation) loss:

$$\mathcal{L}_{\text{global}} = \frac{1}{N} \sum_{i=1}^N \left[ -\log \frac{\exp(S_{\text{global},i})}{\sum_j \exp(S_{\text{global},j})} \right], \quad (15)$$

where  $N$  is the batch size. This means that the similarity of correctly matched video text pairs in  $S_{\text{global}}$  will be maximized, while the similarity of mismatched sample pairs will be minimized, making the features of different categories more distinguishable.

For local frame and word alignment, we compute a frame-word similarity matrix  $S_{\text{local}} \in \mathbb{R}^{N_F \times N_T}$  using Einstein summation:

$$S_{\text{local}}[t, s] = \tau \langle v_{\text{series}}^t, t_{\text{series}}^s \rangle, \quad (16)$$

$$\text{for } t = 1, \dots, N_F, \text{ and } s = 1, \dots, N_T, \quad (17)$$

where  $v_{\text{series}}^t \in \mathbb{R}^D$  is the feature vector of the  $t$ th video frame, and  $t_{\text{series}}^s \in \mathbb{R}^D$  is the feature vector of the  $s$ th text token, and  $\langle \cdot \rangle$  refers to Einstein summation.

However, not all video frames and text tokens have the same semantic importance. For example, some frames may be crucial for expressing

actions, while others may be redundant or irrelevant. Therefore, a time-sensitive weight  $w_{time}$  is needed to control which frames contribute more in contrastive learning. To emphasize temporal dynamics, we derive time-sensitive weights by applying the softmax function to the maximum similarity across tokens for each frame:

$$w_{time}[t] = \text{softmax}\left(\max_s S_{local}[t, s]\right). \quad (18)$$

The positive alignment score, representing the correspondence of aligned frame-word pairs, is then aggregated as

$$S_{pos} = \sum_{i=1}^{N_T} \text{diag}(S_{local})_i \cdot w_{time}[t]. \quad (19)$$

To compute the negative sample similarity, we take the mean similarity score over all non-matching pairs within the batch. Specifically, the negative similarity score  $S_{neg}$  is formulated as:

$$S_{neg} = \frac{1}{(N-1)} \sum_{i \neq j} S_{local}^{(i,j)}, \quad (20)$$

where  $S_{local}^{(i,j)}$  represents the local similarity score between video sample  $i$  and text sample  $j$  within the batch. Since all non-matching samples in the batch serve as negative samples in contrastive learning, we average over all off-diagonal elements to ensure that the learned representations are not biased toward a single positive pair but instead leverage the full distribution of negative samples.

To construct the **local timing alignment loss**, we adopt the InfoNCE loss, which is defined as:

$$\mathcal{L}_{local} = -\log \frac{\exp(S_{pos})}{\exp(S_{pos}) + \sum_{k \neq i} \exp(S_{neg}^{(k)})} \quad (21)$$

where  $\exp(S_{pos})$  in the numerator represents the similarity score of the positive sample pair, while the denominator aggregates the similarity scores of all negative samples. The softmax normalization ensures that the model effectively learns to distinguish positive pairs from negative ones, thereby reinforcing temporal alignment in local feature representations.

The hierarchical contrastive loss combines the local contrastive loss  $\mathcal{L}_{local}$  with the global contrastive loss  $\mathcal{L}_{global}$  to ensure optimal alignment at both global and temporal levels:

$$\mathcal{L}_{align} = \lambda_{global} \mathcal{L}_{global} + \lambda_{local} \mathcal{L}_{local}, \quad (22)$$

where  $\lambda_{global}$  and  $\lambda_{local}$  are tunable hyperparameters that balance the contributions of global feature alignment and local temporal alignment, which is further analyzed in Section 3.6.

The aligned video global feature  $v_{global}$  is then regressed to the complexity prediction of the motion branch:

$$P_M = \text{Linear}(v_{global}), \quad (23)$$

$$L_M = \frac{1}{N} \sum_{i \in N} (P_M^i - S_I^i)^2. \quad (24)$$

And the final loss is computed as:

$$\alpha L_I + \beta L_M + \text{Normalize}(\mathcal{L}_{align}), \quad (25)$$

where  $\text{Normalize}(\cdot)$  is applied to ensure that  $\mathcal{L}_{align}$  remains on the same scale as  $L_I$  and  $L_M$ , preventing excessive alignment loss from interfering with the learning of complexity levels.

### 3. Experimental results and analysis

#### 3.1. Dataset and evaluation metric

We conducted experiments on three publicly available image complexity assessment datasets, including the IC9600 dataset [12], VISC-C [63], and Savoias [64]. The IC9600 dataset is a publicly available,

large-scale image complexity assessment dataset. It contains 9600 images spanning eight semantic categories: abstract, advertisement, architecture, objects, paintings, people, scenes, and transportation. Each image was annotated by 17 rigorously trained annotators to minimize individual bias. The ‘‘Vischema-Complexity’’ (VISC-C) dataset is based on the VISHEMA categorical scene dataset [65] and contains 800 images with 800 corresponding complexity scores, each image rated by human observers. Savoias is a visual complexity dataset comprising over 1400 images from seven categories relevant to the research areas discussed above: Scenes, Advertisements, Visualization and infographics, Objects, Interior design, Art, and Suprematism. The images within each category display diverse characteristics, including various low-level and high-level features, objects, backgrounds, textures, patterns, text, and graphics. Ground truth for Savoias is obtained through crowdsourcing over 37,000 pairwise comparisons of images using the forced-choice methodology and involving more than 1600 contributors. The resulting relative scores are then converted to absolute visual complexity scores using the Bradley-Terry method and matrix completion.

Following the evaluation protocol of ICNet [12], we adopt four metrics to assess model performance, including Pearson Linear Correlation Coefficient (PLCC), Spearman Rank-Order Correlation Coefficient (SRCC), Root Mean Squared Error (RMSE), and Kendall Rank-Order Correlation Coefficient (KRCC). PLCC measures the linear correlation between predicted and subjective scores, while SRCC evaluates their rank correlation. RMSE quantifies the prediction errors, which focus on the squared differences. KRCC is used to measure the ordinal association between two measured quantities. An ideal quality metric would have SRCC, KRCC, and PLCC values close to one, and RMSE close to zero. Together, these metrics comprehensively evaluate the model’s accuracy and consistency in predicting image complexity.

#### 3.2. Implement details

The datasets are randomly divided into training and testing sets with an 8:2 ratio. During training, the batch size per iteration is 32, and the model is optimized using the SGD algorithm with a learning rate of 0.05 over 30 epochs. Empirically, the hyperparameters  $\alpha$  and  $\beta$  are set to 0.8 and 0.2, respectively. And the  $\lambda_{global}$  and  $\lambda_{local}$  are set to 0.6 and 0.4, respectively. Further ablation studies about the hyperparameters, variants of image/video encoders, and I2V/I2T pretrained models are detailed in the following sections. All experiments are implemented in PyTorch on a 24 GB 3090 GPU.

#### 3.3. Performance comparisons

We first compare our proposed method with state-of-the-art ICA models on the IC9600 dataset. We also compare with five other advanced image quality assessment (IQA) metrics to ensure our effectiveness. Specifically, the ICA methods include CNet [66] and ICNet [12]. CNet generated both single-score complexity ratings and two-dimensional per-pixel complexity maps. It integrates scene clutter and symmetry metrics to enhance neural network performance. ICNet introduced a dual-branch architecture, combining high-resolution detail and downsampled context features. The IQA methods include DBCNN [67], NIMA [68], HyperIQA [69], CLIPQA [70] and TOPIQ [30].

As shown in Table 1, our proposed model achieves the highest correlation scores across all key metrics (SRCC = 0.943, KRCC = 0.812, PLCC = 0.953), surpassing TOPIQ (0.938, 0.799, 0.944) and ICNet (0.937, 0.797, 0.946). To further validate its robustness, we evaluate our method on the VISC-C and Savoias datasets, with results presented in Tables 2 and 3.

On VISC-C (Table 2), OURS outperforms all models, achieving SRCC (0.851), KRCC (0.654), PLCC (0.829), and RMSE (0.060). Compared to ICNet, the second-best ICA model, OURS improves SRCC by 3.3%,

**Table 1**

Performance comparisons on the IC9600 dataset. The best-performing results are highlighted in bold. The calculations of model Flops are based on the input image shape  $3 \times 512 \times 512$ , and the inference speed is taken as an average of 100 inferences. Note that the calculations of model parameters and inference speed for our proposed method consist of three parts – base, I2T, and I2V, where the base refers to the main body of the model, excluding I2T and I2V. And the utilized I2T and I2V models are the InstructBlip-7B and CogVideoX, respectively.

| Model |                 | Year | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        | Params/M     | Flops/G | Speed      |
|-------|-----------------|------|--------------|--------------|--------------|--------------|--------------|---------|------------|
| IQA   | DBCNN [67]      | 2018 | 0.871        | 0.699        | 0.879        | 0.071        | 15.311       | 86.219  | 23.688 ms  |
|       | NIMA [68]       | 2018 | 0.838        | 0.688        | 0.555        | 0.194        | 54.322       | 13.181  | 25.001 ms  |
|       | HyperIQA [69]   | 2020 | 0.926        | 0.780        | 0.933        | 0.204        | 27.375       | 107.831 | 29.638 ms  |
|       | CLIPQA [70]     | 2023 | 0.897        | 0.735        | 0.898        | 0.078        | –            | 61.065  | 21.755 ms  |
|       | TOPIQ [30]      | 2024 | 0.938        | 0.799        | 0.944        | 0.049        | 45.197       | 37.261  | 14.434 ms  |
| ICA   | CNet [66]       | 2022 | 0.870        | –            | 0.873        | –            | –            | –       | –          |
|       | ICNet [12]      | 2023 | 0.937        | 0.797        | 0.946        | <b>0.049</b> | 20.328       | 28.399  | 4.563 ms   |
|       | OURS (base +I2T | 2025 | <b>0.943</b> | <b>0.812</b> | <b>0.953</b> | 0.060        | (173.462     | 41.836  | (17.392 ms |
|       | +I2V)           |      |              |              |              |              | +446.290     | –       | +0.6503 s  |
|       |                 |      |              |              |              |              | +10 548.571) | –       | +~180 s)   |

**Table 2**

Performance comparisons on the VISC-C dataset. The best-performing results are highlighted in bold.

| Model |          | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
|-------|----------|--------------|--------------|--------------|--------------|
| IQA   | DBCNN    | 0.779        | 0.582        | 0.783        | 0.086        |
|       | NIMA     | 0.810        | 0.611        | 0.803        | 0.125        |
|       | HyperIQA | 0.734        | 0.533        | 0.739        | 0.181        |
|       | CLIPQA   | 0.781        | 0.597        | 0.796        | 0.122        |
|       | TOPIQ    | 0.803        | 0.610        | 0.811        | 0.079        |
| ICA   | CNet     | –            | –            | 0.716        | –            |
|       | ICNet    | 0.818        | 0.619        | 0.814        | 0.079        |
|       | OURS     | <b>0.851</b> | <b>0.654</b> | <b>0.829</b> | <b>0.060</b> |

**Table 3**

Performance comparisons on the Savoias dataset. The best-performing results are highlighted in bold.

| Model |          | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
|-------|----------|--------------|--------------|--------------|--------------|
| IQA   | DBCNN    | 0.768        | 0.575        | 0.770        | 0.147        |
|       | NIMA     | 0.781        | 0.584        | 0.771        | 0.210        |
|       | HyperIQA | 0.801        | 0.612        | 0.798        | 0.293        |
|       | CLIPQA   | 0.779        | 0.594        | 0.794        | 0.101        |
|       | TOPIQ    | 0.838        | 0.648        | 0.832        | 0.123        |
| ICA   | ICNet    | 0.865        | 0.675        | 0.849        | 0.121        |
|       | OURS     | <b>0.887</b> | <b>0.705</b> | <b>0.879</b> | <b>0.092</b> |

KRCC by 3.5%, PLCC by 1.5%, and reduces RMSE by 24.1%, highlighting its significant performance gain. Additionally, OURS surpasses all IQA models, demonstrating the advantage of integrating motion-related features for image complexity assessment.

Similarly, on Savoias (Table 3), OURS maintains its superiority with SRCC (0.887), KRCC (0.705), PLCC (0.879), and RMSE (0.092). Compared to ICNet, OURS improves SRCC by 2.5%, KRCC by 4.4%, PLCC by 3.5%, and reduces RMSE by 23.8%, further demonstrating its effectiveness across diverse datasets.

These results confirm that OURS consistently surpasses state-of-the-art ICA and IQA models, exhibiting stronger correlation with human perception and higher reliability in the ICA task. The performance gap underscores the effectiveness of leveraging motion-aware representations, making OURS a promising solution for future image assessment tasks.

### 3.4. Cross-dataset validation

Table 4 presents the results of cross-dataset validation, comparing IQA and ICA models across different training and testing datasets. Our proposed model consistently achieves the highest SRCC and PLCC scores in all test scenarios, demonstrating superior generalization. ICA models (OURS and ICNet) outperform IQA methods, with OURS achieving the best results. IQA methods like HyperIQA and CLIPQA perform well but fall short of ICA approaches. Overall, OURS exhibits the strongest cross-dataset adaptability and robustness.

### 3.5. Validity on different I2T and I2V models

We have verified the impact of different image-to-text (I2T) and image-to-video (I2V) generation models on performance. This section addresses two main issues. First, does the choice of generation model — yielding varying text prompts and videos — result in significant performance differences? Second, given that different pre-trained generation models are built upon distinct knowledge bases, is there a risk that the text and video outputs might introduce extraneous information not originally contained in the input image, thereby affecting the evaluation metrics?

For image-to-text (I2T) models, following the protocols of LAVIS [71], the pre-trained BLIP [57] model is fine-tuned on the large COCO dataset. The BLIP-2 [58] model utilizes the OPT-2.7B language model and is fine-tuned on COCO. In accordance with the InstructBLIP-XL [59] framework, it employs the Flan-T5-XL language model, which has been instruction-tuned on a mixture of datasets including C4 and various task-specific datasets. Similarly, the InstructBLIP-7B model incorporates the Vicuna-7B language model, which is fine-tuned from LLaMA [72]. Despite differences in pretraining data, the results presented in Table 5 indicate minimal performance variations across these I2T models on the IC9600 and VISC-C datasets.

For image-to-video (I2V) models, according to the protocols of CogVideoX [25] and I2VGen-XL [60], the CogVideoX1.5 and CogVideoX models are trained on their self-collected high-quality video clips with generated text descriptions. The I2VGen-XL model is trained on both public datasets, which comprise WebVid10M [73] and LAION-400M [74], as well as private datasets consisting of video-text pairs and image-text pairs of the same type. As shown in Table 6, the performance differences among these I2V variants are also insignificant. In summary, the pre-trained I2V and I2T models based on various training data do not significantly influence the effectiveness of the proposed model.

### 3.6. Ablation studies on hyperparameters

Table 7 presents an ablation study of  $\alpha$  and  $\beta$ . The experiments reveal that the weight distribution in the dual-branch structure significantly impacts model performance. When  $\alpha = 0.8$  and  $\beta = 0.2$ , the model achieves optimal performance with SRCC (0.943), KRCC (0.812), PLCC (0.953), and RMSE (0.035), suggesting that the image branch (dominated by  $\alpha$ ) should carry a higher weight. Still, it must be moderately combined with the motion branch ( $\alpha = 0.2$ ) to retain essential information. When  $\alpha \leq 0.6$ , performance consistently deteriorates with decreasing  $\alpha$  (e.g., SRCC drops to 0.902 when  $\alpha = 0.0$ ), demonstrating that dominance by a single branch can severely undermine the model's capability. Therefore, balancing the contributions of both branches is key to enhancing the model's robustness.

Table 8 shows the ablation study on  $\lambda_{\text{global}}$  and  $\lambda_{\text{local}}$ . When  $\lambda_{\text{global}} = 0.6$  and  $\lambda_{\text{local}} = 0.4$ , the model performs best (SRCC = 0.943, RMSE = 0.035), indicating that global alignment is more critical for modeling

**Table 4**

Cross-dataset validation. The best performances are highlighted in bold.

| Train |          | IC9600       |              |              |              | VISC-C       |              |              |              | Savoias      |              |              |              |
|-------|----------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Test  |          | VISC-C       |              | Savoias      |              | IC9600       |              | Savoias      |              | IC9600       |              | VISC-C       |              |
| Model |          | SRCC↑        | PLCC↑        | SRCC↑        | PLCC↑        | SRCC         | PLCC         | SRCC↑        | PLCC↑        | SRCC↑        | PLCC↑        | SRCC↑        | PLCC↑        |
| IQA   | DBCNN    | 0.685        | 0.674        | 0.606        | 0.626        | 0.742        | 0.750        | 0.525        | 0.538        | 0.773        | 0.787        | 0.644        | 0.652        |
|       | NIMA     | 0.566        | 0.297        | 0.603        | 0.240        | 0.714        | 0.390        | 0.575        | 0.540        | 0.818        | 0.725        | 0.630        | 0.588        |
|       | HyperIQA | 0.711        | 0.688        | 0.669        | 0.669        | 0.668        | 0.668        | 0.550        | 0.554        | 0.761        | 0.748        | 0.643        | 0.629        |
|       | CLIPQA   | 0.702        | 0.686        | 0.600        | 0.608        | 0.406        | 0.399        | 0.487        | 0.481        | 0.789        | 0.775        | 0.550        | 0.568        |
|       | TOPIQ    | 0.759        | 0.730        | 0.659        | 0.648        | 0.695        | 0.700        | 0.594        | 0.597        | 0.806        | 0.804        | 0.674        | 0.685        |
| ICA   | ICNet    | 0.753        | 0.712        | 0.728        | 0.716        | 0.718        | 0.739        | 0.664        | 0.660        | 0.805        | 0.807        | 0.712        | 0.709        |
|       | OURS     | <b>0.761</b> | <b>0.736</b> | <b>0.740</b> | <b>0.745</b> | <b>0.815</b> | <b>0.818</b> | <b>0.680</b> | <b>0.673</b> | <b>0.818</b> | <b>0.823</b> | <b>0.725</b> | <b>0.721</b> |

**Table 5**

Validation on image-to-text (I2T) generation models. Four state-of-the-art I2T models are compared. The I2V model utilized in this validation is the CogVideoX [25]. The best results are highlighted in bold.

| Dataset              | IC9600       |              |              |              | VISC-C       |              |              |              |
|----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                      | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
| Blip [57]            | 0.939        | 0.802        | 0.946        | 0.038        | 0.846        | <b>0.655</b> | 0.828        | <b>0.058</b> |
| Blip2 [58]           | 0.940        | 0.804        | 0.947        | 0.038        | 0.846        | 0.649        | 0.821        | 0.076        |
| InstructBlip-XL [59] | 0.938        | 0.801        | 0.946        | 0.038        | 0.843        | 0.642        | 0.816        | 0.065        |
| InstructBlip-7B [59] | <b>0.943</b> | <b>0.812</b> | <b>0.953</b> | <b>0.035</b> | <b>0.851</b> | 0.654        | <b>0.829</b> | 0.060        |

**Table 6**

Validation on image-to-video (I2V) generation models. Three state-of-the-art I2T models are compared. The I2T model utilized in this validation is the InstructBLIP-7B [59]. The best results are highlighted in bold.

| Dataset           | VISC-C       |              |              |              | Savoias      |              |              |              |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        | SRCC↑        | KRCC↑        | PLCC↑        | RMSE ↓       |
| CogVideoX1.5 [25] | 0.847        | <b>0.656</b> | 0.825        | 0.060        | 0.882        | 0.702        | 0.873        | 0.092        |
| CogVideoX [25]    | <b>0.851</b> | 0.654        | <b>0.829</b> | <b>0.060</b> | <b>0.887</b> | <b>0.705</b> | <b>0.879</b> | 0.092        |
| I2VGen-XL [60]    | 0.834        | 0.644        | 0.827        | 0.068        | 0.878        | 0.698        | 0.869        | <b>0.090</b> |

**Table 7**Ablation studies of the hyperparameters –  $\alpha$  and  $\beta$  on IC9600 dataset, which decide which branch would dominate in the dual branch structure. The best results are highlighted in bold.

| $\alpha$ | $\beta$ | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
|----------|---------|--------------|--------------|--------------|--------------|
| 1.0      | 0.0     | 0.930        | 0.786        | 0.940        | 0.047        |
| 0.8      | 0.2     | <b>0.943</b> | <b>0.812</b> | <b>0.953</b> | <b>0.035</b> |
| 0.6      | 0.4     | 0.922        | 0.775        | 0.932        | 0.066        |
| 0.4      | 0.6     | 0.918        | 0.767        | 0.920        | 0.047        |
| 0.2      | 0.8     | 0.910        | 0.754        | 0.910        | 0.053        |
| 0.0      | 1.0     | 0.902        | 0.743        | 0.897        | 0.051        |

**Table 8**Ablation studies of the hyperparameters –  $\lambda_{\text{global}}$  and  $\lambda_{\text{local}}$  on IC9600 dataset, which emphasize the significance of both global feature alignment and local (i.e., frame and word) level alignment in the video-to-text alignment process, highlighting that the hierarchical loss function effectively guides the model to prioritize the extraction of motion information. The best results are highlighted in bold.

| $\lambda_{\text{global}}$ | $\lambda_{\text{local}}$ | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
|---------------------------|--------------------------|--------------|--------------|--------------|--------------|
| 0.0                       | 1.0                      | 0.938        | 0.801        | 0.931        | 0.056        |
| 0.2                       | 0.8                      | 0.940        | 0.806        | 0.951        | 0.047        |
| 0.4                       | 0.6                      | 0.939        | 0.805        | 0.952        | 0.046        |
| 0.6                       | 0.4                      | <b>0.943</b> | <b>0.812</b> | <b>0.954</b> | <b>0.035</b> |
| 0.8                       | 0.2                      | 0.940        | 0.807        | 0.951        | 0.037        |
| 1.0                       | 0.0                      | 0.939        | 0.804        | 0.949        | 0.037        |

motion information, yet local (i.e., frame and word) alignment is still necessary to supplement detailed features. Extreme configurations (such as  $\lambda_{\text{global}} = 1.0$  or  $\lambda_{\text{local}} = 1.0$ ) result in a drop in performance metrics; for instance, relying solely on local alignment ( $\lambda_{\text{global}} = 0.0$ ) increases the RMSE to 0.056. This demonstrates that the hierarchical loss, by dynamically balancing global and local features, effectively guides the model to focus on extracting information related to motion cues while avoiding overfitting to local noise.

### 3.7. Ablation studies on motion branch

Table 9 presents an ablation study on the motion branch, evaluating the impact of different components on performance. On both the VISC-C and Savoias datasets, removing cross-modal attention (“w/o Atten”) and hierarchical loss (“w/o  $L_H$ ”) leads to a performance drop, highlighting their importance in video-text alignment and regression. Using only text features for prediction (“Solo Text”) performs significantly worse than models incorporating video features, demonstrating the critical role of video information in motion modeling. Adding extra text features (“Add Text”) does not provide a noticeable performance gain, indicating that video features are more effective for motion measurements. Ultimately, the complete model (“OURS”) achieves the best results across most metrics, validating the effectiveness of the proposed approach.

### 3.8. Validation on further image/video encoders

Table 10 presents the validation results of different image and video encoders on the Savoias and IC9600 datasets. Three image encoders (ResNet34 [56], ResNet101, and ViT [75]) and three video encoders (CNN+GRU, C3D [76], and TimeSFormer [77]) are evaluated based on SRCC, KRCC, PLCC, and RMSE metrics. The results indicate that although ViT combined with CNN+GRU achieves the highest SRCC (0.999) and KRCC (0.803) on IC9600, the performance gap between various image encoders is minimal. Among the video encoders, TimeSFormer achieves the best performance on the Savoias dataset with an SRCC of 0.888 and a KRCC of 0.711. However, the choice of image and video encoders will not affect the model performance on these ICA datasets.





Fig. 6. The visualizations of image complexity predictions. The images are sampled from the IC9600 dataset. The MOS/Prediction with alignment/Prediction without alignment for each image is rounded to 3 decimal places for convenience. The “alignment” refers to video-text alignment.

Table 9

Ablation studies on the motion branch. “w/o Atten”: Removes the cross-modal attention mechanism in video-to-text alignment. “w/o  $L_H$ ”: Eliminates the hierarchical loss  $L_{Align}$ . “Solo Text”: Only text features are used for prediction in the motion branch, without incorporating video features. “Add Text”: Integrates the aligned text features  $t_{global}$  with video features  $v_{global}$  for motion branch regression. The best results are highlighted in bold.

| Dataset   | VSIC-C       |              |              |              | Savoias      |              |              |              |
|-----------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Model     | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
| w/o Atten | 0.802        | 0.597        | 0.793        | 0.069        | 0.847        | 0.655        | 0.837        | 0.107        |
| w/o $L_H$ | 0.814        | 0.616        | 0.801        | 0.066        | 0.864        | 0.687        | 0.858        | 0.099        |
| Solo Text | 0.808        | 0.610        | 0.800        | 0.064        | 0.852        | 0.665        | 0.843        | 0.106        |
| Add Text  | 0.834        | 0.642        | 0.823        | <b>0.057</b> | 0.885        | 0.707        | 0.872        | <b>0.089</b> |
| OURS      | <b>0.851</b> | <b>0.654</b> | <b>0.829</b> | 0.060        | <b>0.887</b> | <b>0.705</b> | <b>0.879</b> | 0.092        |

Table 10

Validation on Image/Video Encoders. We evaluate the impact of different encoders on final performance by utilizing three image encoders and three video encoders. The best results are highlighted in bold.

| Encoder   |                  | Savoias      |              |              |              | IC9600       |              |              |              |
|-----------|------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Image     | Video            | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
| ResNet34  | CNN+GRU          | 0.887        | 0.705        | <b>0.879</b> | <b>0.092</b> | 0.943        | 0.812        | 0.953        | <b>0.035</b> |
| ResNet101 | CNN+GRU          | 0.872        | 0.685        | 0.863        | 0.092        | <b>0.946</b> | <b>0.814</b> | <b>0.954</b> | 0.037        |
| VIT [75]  | CNN+GRU          | 0.870        | 0.686        | 0.865        | 0.092        | 0.939        | 0.803        | 0.946        | 0.041        |
| ResNet34  | C3D [76]         | 0.885        | 0.707        | 0.873        | 0.096        | 0.944        | 0.813        | 0.954        | 0.042        |
| ResNet34  | TimeSFormer [77] | <b>0.888</b> | <b>0.711</b> | 0.875        | 0.097        | 0.943        | 0.812        | 0.953        | 0.041        |

Table 11

The computational complexity and inference details of the image-to-video (I2V) models.

| I2V models   | Video resolution | Number of frames | Model parameters/M | Inference speed (Step = 32) | Video length | Frame rate |
|--------------|------------------|------------------|--------------------|-----------------------------|--------------|------------|
| CogVideoX1.5 | 512 × 512        | 24               | 10 548.571         | ~117 S                      | ~3 s         | 8          |
| CogVideoX    | 720 × 480        | 49               | 10 548.177         | ~180 S                      | ~2 s         | 24         |

Table 12

Validation on the temperature parameter  $\tau$  on both IC9600 and VISC-C datasets. The best results are highlighted in bold.

| Param | IC9600       |              |              |              | VISC-C       |              |              |              |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|       | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        | SRCC↑        | KRCC↑        | PLCC↑        | RMSE↓        |
| 0.03  | 0.910        | 0.708        | 0.921        | 0.058        | 0.844        | 0.648        | 0.819        | 0.066        |
| 0.04  | 0.922        | 0.777        | 0.936        | 0.046        | 0.846        | 0.658        | 0.831        | 0.065        |
| 0.05  | 0.924        | 0.780        | 0.937        | 0.044        | 0.845        | 0.653        | 0.826        | 0.067        |
| 0.06  | 0.940        | 0.951        | 0.951        | 0.047        | 0.852        | 0.657        | 0.830        | 0.066        |
| 0.07  | <b>0.943</b> | <b>0.812</b> | <b>0.953</b> | <b>0.035</b> | 0.851        | 0.654        | 0.829        | <b>0.060</b> |
| 0.08  | 0.942        | 0.811        | 0.953        | 0.057        | 0.854        | 0.661        | 0.831        | 0.065        |
| 0.09  | 0.941        | 0.808        | 0.952        | 0.057        | <b>0.861</b> | <b>0.672</b> | <b>0.836</b> | 0.064        |
| 0.10  | 0.942        | 0.809        | 0.952        | 0.059        | 0.842        | 0.649        | 0.825        | 0.067        |

### 3.9. Validity of the temperature parameter

As shown in Table 12, the temperature parameter  $\tau$  significantly impacts performance. As  $\tau$  increases from 0.03 to approximately 0.07–0.09, the model demonstrates consistent improvements across metrics such as SRCC and PLCC on both the IC9600 and VISC-C datasets, reaching optimal performance within this range and slightly declining thereafter. In contrastive learning, the temperature parameter controls the sharpness of the softmax distribution, thereby influencing the relative contribution of positive and negative samples to the gradient. A lower temperature (e.g.,  $\tau = 0.03$ ) sharpens the distribution, amplifying the influence of hard negatives, which can be beneficial in challenging

settings but may lead to overfitting to a limited subset of difficult examples. In contrast, moderate temperatures (e.g.,  $\tau = 0.07$  to  $\tau = 0.09$ ) promote a more balanced gradient contribution from both positives and negatives, which is particularly suitable for the current task scenario where the majority of text-video pairs exhibit high semantic alignment and hard negatives are relatively rare. Since most texts correspond closely to their associated videos, extreme emphasis on difficult negatives is unnecessary.

### 3.10. Visualization of complexity predictions

The results presented in Fig. 6 showcase the images for which the predictions of the model variant without the cross-modal alignment in the motion branch deviate more significantly from the ground-truth MOS. Notably, these images exhibit strong implicit motion cues, further supporting our claim that cross-modal alignment effectively enhances the sensitivity of video features to such motion-related signals.

### 3.11. Limitations

Note that the additional capture of motion information brings about greater model complexity. As shown in Table 1, we provide a detailed analysis of the computational complexity associated with the pretrained I2T and I2V models. The “base” model of our method refers to the core framework without incorporating these pretrained models. While our approach achieves superior performance across all evaluation metrics,

it is worth noting that the I2T and I2V (as detailed in Table 11) components significantly increase computational cost, both in terms of parameters and inference time, which may hinder our applicability in real-time scenarios. In future work, we aim to explore more lightweight and efficient end-to-end alternatives to further enhance processing speed while maintaining high-quality motion information extraction.

#### 4. Conclusion

In this paper, we introduced a novel approach to image complexity assessment by unlocking implicit motion dynamics inspired by the inherent dynamic mechanism of the human visual system. Our motion-inspired image complexity assessment metric (MICM) method employs a dual-branch architecture that extracts spatial features from static images while simultaneously generating short video sequences to capture motion information explicitly. We designed a hierarchical loss function that refines motion semantics through text-based supervision to enhance the alignment between motion representation and semantic meaning. Extensive experiments on three public ICA datasets demonstrate that our approach significantly outperforms advanced ICA and IQA methods, validating its effectiveness in capturing the nuanced nature of image complexity. This work provides new insights into leveraging implicit motion for visual perception tasks.

#### CRedit authorship contribution statement

**Yixiao Li:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Xiaoyuan Yang:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Yuqing Luo:** Methodology, Writing – original draft. **Hadi Amirpour:** Formal analysis, Methodology, Writing – review & editing. **Hantao Liu:** Investigation, Methodology, Software, Writing – review & editing. **Wei Zhou:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Data availability

Data will be made available on request.

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